ISEN/DAEN 427 Lecture 6.1
Mastering the Metropolis Algorithm
A concise guide

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based on Sarah Lee, Number Analytics

What is the Metropolis Algorithm?

- A foundational Markov Chain Monte Carlo (MCMC) method for sampling from complex probability distributions.
- Origins: introduced in Metropolis et al., 1953; later generalized by Hastings (1970).
- Widely used in statistical mechanics, computational chemistry, biophysics, and Bayesian computation.

Core components

Proposal distribution Suggests a candidate new state (e.g., Gaussian perturbation).

Acceptance criterion Metropolis–Hastings ratio decides accept/reject to preserve the target distribution.

Ergodicity & Detailed balance Required for convergence: $\pi(x)T(x \to x') = \pi(x')T(x' \to x)$.

Metropolis algorithm (high level)

Input: Initial state x_0, target density $\pi(x)$ known up to constant, iterations N.

for $t \leftarrow 1$ **to** N **do**

Propose $x' \sim q(x'|x_{t-1})$;

Compute acceptance ratio

$$a = \min\left(1, \frac{\pi(x')\, q(x_{t-1}|x')}{\pi(x_{t-1})\, q(x'|x_{t-1})}\right)$$

;

With probability a set $x_t \leftarrow x'$, else $x_t \leftarrow x_{t-1}$;

end

Algorithm 1: Metropolis(-Hastings) sampler

Choosing the proposal distribution

- Symmetric proposal (e.g., Gaussian centered at current state) simplifies the ratio since $q(x'|x) = q(x|x')$.
- Typical acceptance target: roughly 20%–50% depending on dimensionality and problem. [Reference 6]
- Trade-off: small steps $\Rightarrow$ high acceptance but slow exploration; large steps $\Rightarrow$ low acceptance and wasted proposals.

Convergence diagnostics (brief)

- Monitor trace plots and running means.
- Compute effective sample size (ESS) and autocorrelation.
- Run multiple chains from dispersed starting points to check for consistent stationary behavior.

Applications in science

- Ising model simulations and phase transitions (study critical phenomena).
- Protein folding / energy landscape exploration in biophysics.
- General condensed matter problems (disordered systems, superconductors, superfluids).
- Optimization variants: simulated annealing (temperature schedule to seek global minima).

Implementation tips & pitfalls

- Ensure ergodicity: design proposals so the chain can reach the full state space.
- Maintain detailed balance (or otherwise correct for asymmetry using Hastings adjustment).
- Watch out for poor mixing in high dimensions — consider adaptive proposals or more advanced MCMC (Hamiltonian Monte Carlo, Gibbs sampling where applicable).
- Diagnose convergence; do not rely solely on a single long chain.

Conclusion

- The Metropolis algorithm remains a simple, robust MCMC workhorse for sampling complex distributions and simulating statistical -mechanics systems.
- Choice of proposal and convergence diagnostics are key to practical success.

References

- N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, "Equation of State Calculations by Fast Computing Machines," *Journal of Chemical Physics*, 1953.
- W. K. Hastings, "Monte Carlo Sampling Methods Using Markov Chains and Their Applications," *Biometrika*, 1970.
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